

CyberArk Privileged Access Security

Azure Connector Estate · Post-Migration Test and Validation Plan

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Application	CyberArk PAS, Privilege Cloud connectors	Application owner	Eric Ullmann, GEHA Security
Scope	Production connector pair, parallel run; disaster recovery against the DR edition when issued	Peer reviewer	Bibbin Joseph
Basis	Azure Future State Build Tables v2.3 Production; Production Build Handover and Address Register v2.3	Status	Draft

Summary

What is being tested. Two new CyberArk Privilege Cloud connectors in Azure, sitting behind an internal load balancer at cyberark-prd.geha.com. Nothing is moved or switched off. The on-premises connectors keep serving every server that has not migrated, and the vault, portal and discovery stay in the CyberArk tenant as they are today. This is a parallel run, and the plan is built around proving the two estates coexist safely.

How the test is organized. Seven groups over four days. Day 1 confirms the Azure build matches the build tables (group A), every network path opens and the deny path refuses (group B), and both connectors register cleanly in the tenant beside the on-premises pair (group C). Days 2 and 3 put real sessions through the Azure connectors, spread them across both servers, and take one server away to prove failover (group D). Day 3 also runs the controls that keep the two estates apart (group E) and checks that backup, alerting, patching and the CyberArk agent upgrade windows are in place (group F). Day 4 proves the connectors replicate to East US 2 and can be started there on a test failover (group G); it runs against the disaster recovery edition of the build tables when that edition is issued.

The two tests that matter most. Group E moves an account so that an Azure connector is asked to reach a server that has not migrated; the expected result is a refusal logged at the hub firewall against the deny rule. If that session ever reaches the target, testing stops and no cutover request goes forward until the rule is fixed and the case reruns clean. The second is the failover pair in group D: stop the session manager on one connector, then take a whole availability zone away, and confirm users on the other connector never notice.

How a case passes. Every case names the steps, the role that runs it, the evidence file to save and the expected result. A case passes only when that evidence is in the evidence store and matches. Anything without evidence is recorded as not run, never as passed. The pack goes to GEHA Security and Enterprise Architecture for acceptance before the CAB agenda request.

Who is involved. A build engineer from the joint GEHA and CHC build team runs the build and network cases and assembles the evidence. GEHA's CyberArk administrator owns everything in the tenant. The Cloud Platform Team owns the subnet and the platform network; GEHA Network owns the hub firewall policy; GEHA Identity owns the connector organizational unit and the CyberArk hardening GPO. A GEHA test user opens the sessions. Service owners for Splunk, MECM, ServiceNow, RDS licensing, Cohesity, Rapid7, endpoint protection and DNS confirm their services see the new servers. Plan on about 14 hours of hands-on execution over days 1 to 3, roughly 3.8 hours on day 4, plus the built-in waits for configuration refresh, password manager cycles and initial replication.

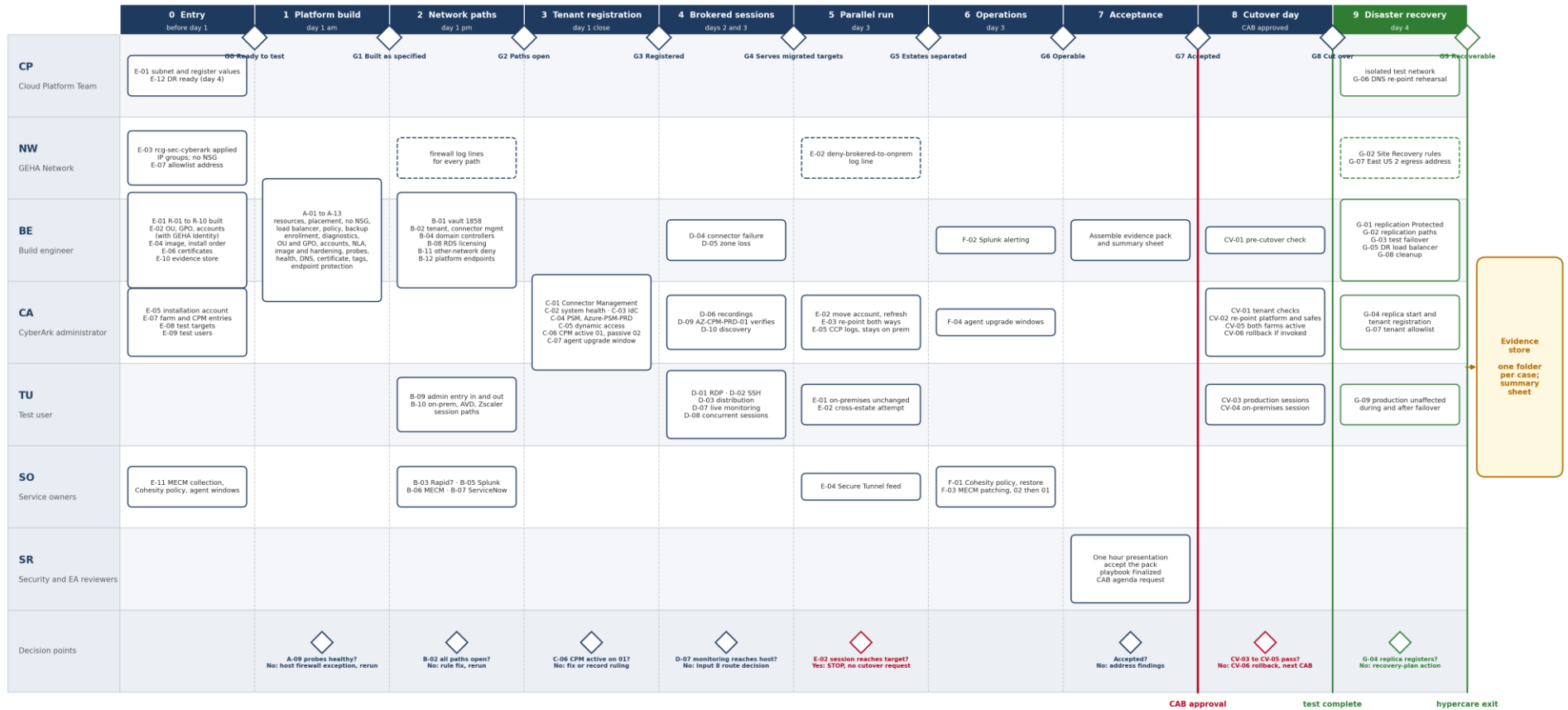
Cutover and rollback. Cutover is a tenant configuration change, not an infrastructure change: the first production platform is pointed at the Azure session manager farm, one session is run each way, and both farms are confirmed active. If anything fails, the platform is pointed back at the on-premises farm and the same checks rerun. Rollback takes under 30 minutes and no system is taken out of service. The rollback call belongs to Phil Costanzo, GEHA's admin and support manager for CyberArk; cutover itself is approved by Security, Enterprise Architecture and CAB together. The Rollback Plan has the full procedure. Cutover is scheduled in a CAB maintenance window, Thursday evening or a weekend, outside the Open Season, Thanksgiving and year-end freezes in the wave plan.

What is still open. The build tables list the inputs GEHA still owes in their inputs table. Five of them shape this plan: confirmation that the password manager runs active on connector 01 and passive on 02; the load balancing method, since CyberArk recommends least-connections routing and a direct route for live session monitoring; the Secure Tunnel audit feed re-point; confirmation that the credential provider tier stays on premises with its consumers; and the automatic upgrade windows for the Management Agent and the Identity Connector. Disaster recovery adds one more: how the tenant treats a replica that starts with the same name as its production connector, which day 4 is designed to find out.

Workflow at a glance

CyberArk PAS · Post-Migration Test and Validation · Workflow v2.1

Lanes are roles; columns are phases; diamonds on the boundaries are gates G0 to G9. Boxes list the cases run in that lane. Every case sends its evidence file to the evidence store.



Waits are built in: 600 s configuration refresh after every platform change (E-02, E-03, CV-02), one CPM cycle (D-09), initial replication before day 4 (G-01). Red rules mark the points where a failure stops the cycle. Basis: Build Tables v2.3 Production.

Validation workflow. Roles as lanes, phases as columns, gates G0 to G9 on the phase boundaries, decision points along the bottom, and the evidence store every case feeds. The workflow description gives the same workflow in table form.

Purpose and scope

This plan validates the two Privilege Cloud connector servers built in subscription CIP-PRD, vm-sec-cyberark-cus-prd-01 and 02 (computer names AZPASPCCPRD101 and AZPASPCCPRD102), in resource group rg-sec-cyberark-prd, behind the internal load balancer lbi-sec-cyberark-cus-prd-01 at cyberark-prd.geha.com, exactly as the Azure Future State Build Tables v2.3 Production describe them. The deployment is parallel: the on-premises connectors PASPCCPRD101 and PASPCCPRD102 keep serving on-premises servers, and the vault, portal and account discovery remain in the Privilege Cloud tenant unchanged.

Sessions are routed to a session manager farm by platform. The Azure pair registers in the tenant as the farm Azure-PSM-PRD whose address is the load balancer name, and a platform sends its accounts to that farm by carrying the farm ID. A migrated server is served by the Azure farm when its account sits on a platform that carries the Azure ID; unmigrated servers stay on platforms that carry the on-premises ID. Credential rotation for migrated servers moves the same way, one safe at a time, to the password manager AZ-CPM-PRD-01, active on connector 01.

The plan proves five things in order: the Azure estate is built as the build tables specify (group A), its network paths work through the hub firewall (B), both connectors are registered in the tenant beside the on-premises pair (C), a migrated server can be brokered through the Azure farm with failover across both zones (D), and the two estates cannot cross (E). Group F confirms the platform services that operate the servers see them and that the operating arrangements in the build tables are in place. Group G proves the connectors replicate to East US 2 and start there on a test failover; it runs against the disaster recovery edition of the build tables when issued, and the names it uses are those of the disaster recovery ruling.

Non-production is validated with the same cases using the non-production names, after production. Performance under production load and the ServiceNow change record are outside this plan. Build tables means Build Tables v2.3 Production and its resource, IP group, firewall rule, load balancer, host configuration, application configuration, operations, tag and inputs tables. Addresses are the register values in those tables until the Cloud Platform Team confirms them.

Roles

Code	Role	Responsibility
BE	Build engineer	GEHA Cloud Platform and CHC Azure build team. Deploys the resources from the Bicep module library, installs the connector software, captures build evidence; on day 4, the Site Recovery objects.
CA	CyberArk administrator	GEHA Security. Owns the Privilege Cloud tenant configuration: session manager farms, platforms, safes, accounts, the tenant allowlist and Connector Management.
CP	Cloud Platform Team	GEHA. Allocates and creates the subnet in the platform group, confirms the register values, owns the landing zone networks and the East US 2 landing zone.
NW	GEHA Network	Owns DefaultFirewallPolicy on the hub firewall; applies the rule collection group rcg-sec-cyberark through the firewall pipeline; pulls firewall log lines.
ID	GEHA Identity	Owns the connector organizational unit, the CyberArk hardening GPO link and the domain accounts in the build tables application configuration.
TU	Test user	A GEHA privileged user working from the administrative ranges, with membership of the test safes.
SO	Service owner	GEHA owner of the platform service under test (Splunk, MECM, ServiceNow, RDS licensing, Cohesity, Rapid7, endpoint protection, DNS).
SR	Security and EA reviewers	GEHA Security and Enterprise Architecture. Accept the evidence pack at the presentation.

Roles used in the case tables.

Entry criteria

Testing starts when every entry criterion is true. Each names who confirms it. E-12 applies to day 4 only.

Ref	Criterion	Detail	Confirmed by
E-01	Build complete	Every Net new resource of the build tables, R-01 to R-10, exists; the subnet was created by the Cloud Platform Team in the platform group; the register values PH-01 to PH-04 either confirmed or deployed as written.	BE, CP
E-02	Directory objects	Both connectors joined to GEHA.com inside the connector organizational unit; the CyberArk Privilege Cloud Connector Unified Hardening GPO linked to that unit and no other hardening policy; the PSMConnect and PSMAdminConnect pair for each connector created per the build tables application configuration.	ID, BE
E-03	Network rules	Rule collection group rcg-sec-cyberark applied to DefaultFirewallPolicy through the firewall pipeline; the net new IP groups of the build tables created with	NW, CP

Ref	Criterion	Detail	Confirmed by
		their members; no network security group on the subnet.	
E-04	Host and connector software	Both servers built from the CIS Level 2 Windows Server 2025 image and configured in the build tables host configuration install order: RDS role, IIS and hosting bundle, connector installer through Connector Management (password manager then session manager), health check component, Identity Connector and DPA connector, Splunk forwarder, endpoint protection agent. Both at the same session manager version; each restarted after the install.	BE, CA
E-05	Installation account	The installation used a domain account with local administrator rights, the Installeruser service user and a Connector Management role per the build tables application configuration; the account used for the second install is not a direct owner of the PSMUnmanagedSessionAccounts safe.	CA
E-06	Certificates	Each connector holds a certificate issued under the GEHA certificate standard whose subject alternative names include the server name and cyberark-prd.geha.com, bound to the HTTPS 443 site.	BE
E-07	Tenant configuration	The session manager farm Azure-PSM-PRD exists as a Configured PSM Server entry, created by copying an existing entry so that PSMProtocolVersion is retained, with Address cyberark-prd.geha.com. The password manager AZ-CPM-PRD-01 is registered. A test platform cloned from the production Windows platform, and one from the Linux platform, carry the Azure farm ID under UI and Workflows, Privileged Session Management. The hub firewall egress address (RE-04) is on the tenant IP allowlist.	CA, NW
E-08	Test targets	One migrated Windows server and one migrated Linux server in the Corp landing zone, each with an account onboarded to a test safe on the matching test platform. One unmigrated on-premises server with an account on a production platform that carries the on-premises farm ID.	CA, SO
E-09	Test users	One test user with the Use Accounts and List Accounts permissions on the test safes and Initiate CPM account management operations; one reviewer with Monitor and View Audit permissions.	CA
E-10	Evidence store	A folder under the program workspace for this test cycle, one subfolder per case, receiving screenshots, exports and logs as each case closes.	BE
E-11	Operations set up	The two connectors are in an MECM maintenance window collection of their own (build tables input 24); the Cohesity protection policy for the Azure connectors is assigned (input 25); the automatic upgrade windows for the Management Agent and the Identity Connector are set or opted out (input 26).	SO, CA
E-12	Disaster recovery ready	Replication enabled on both connectors at least one day before day 4 and both replicated items Protected; the East US 2 landing zone, DR subnet and DR load balancer built per the disaster recovery edition of the build tables; the recovery plan created; an isolated test-failover virtual network in East US 2 provided by the Cloud Platform Team.	BE, CP

Entry criteria.

Test cases

Each case lists the steps in order, the evidence to capture, the expected result, the role that runs it and an estimated duration. A case passes only when the evidence is in the evidence store and matches the expected result. Durations are for one environment and exclude waiting on external teams.

Group A: Platform build verification

Day 1, morning. Proves the Azure estate exists as the build tables specify before any connector test runs. All cases are read only.

Ref	Case	Steps	Evidence	Expected result	Role	Min
A-01	Resource inventory	1. Run az resource list for rg-sec-cyberark-prd and export to CSV. 2. Compare each row with the build tables resource list, R-01 to R-09. 3. Record any resource present but not in the resource list.	CSV export; marked-up copy of the resource list	Every Net new resource present with its name, type, size, zone and location; no extra resources	BE	20
A-02	Network placement	1. Run az network nic show for nic-sec-cyberark-cus-prd-01 and 02; record the subnet reference, private address and accelerated networking. 2. Run az network vnet subnet show for snet-sec-cyberark-cus-prd-01 in rg-alz-centralus-vnet-cip-cus-prd-01; record the prefix, the network security group field and the route table.	Command output for both NICs and the subnet	Both NICs in snet-sec-cyberark-cus-prd-01 at 10.9.250.8 and 10.9.250.9 (or the confirmed register values) with accelerated networking enabled; subnet prefix 10.9.250.0/27; no network security group; route table as set by the platform	BE, CP	15
A-03	Load balancer configuration	1. Open lbi-sec-cyberark-cus-prd-01 in the portal. 2. Record the front end, backend pool members, load balancing rule and both health probes against the build tables load balancer parameters.	Portal screenshots of each blade	Standard regional SKU; zone-redundant static front end 10.9.250.7; both NICs in the pool; rule-rdp TCP 3389 to 3389, no session persistence, idle timeout 20 minutes, TCP reset enabled; probe-rdp TCP 3389 and probe-psm-health HTTPS 443 /psm/api/health, interval 30 seconds, unhealthy threshold 3	BE	15

Ref	Case	Steps	Evidence	Expected result	Role	Min
A-04	Policy compliance and backup enrollment	1. Open Azure Policy, Compliance, filter to rg-sec-cyberark-prd; export the compliance state. 2. Confirm no recovery services vault was created by the Landing Zones backup assignment for these virtual machines.	Compliance export; vault list for the subscription	No non-compliant resource against the Corp and Landing Zones assignments; the backup exclusion tag from build tables input 3 present on both virtual machines and no policy-created vault holds them	BE	15
A-05	Diagnostic settings	1. Open Diagnostic settings on the load balancer and both virtual machines. 2. Record the destination workspace.	Screenshots	Diagnostic settings present on all three resources, sent to alz-log-analytics in the Management subscription	BE	10
A-06	Domain membership, unit and time	1. On each connector run nltest /dsgetdc:geha.com and w32tm /query /status. 2. Run gpresult /r and record the organizational unit and the applied GPOs.	Command output	Secure channel to a GEHA.com domain controller; time source is a domain controller; offset under five minutes; the computer sits in the connector organizational unit; the CyberArk Privilege Cloud Connector Unified Hardening GPO is applied and no other hardening GPO is	BE	15
A-07	Session manager accounts and NLA	1. Open Services on each connector and record the logon account of the CyberArk Privileged Session Manager service and the state of the CyberArk Password Manager service. 2. Open System Properties, Remote, and record the Network Level Authentication setting. 3. Record the members of Remote Desktop Users.	Screenshots	Session manager service runs as that connector's own PSMConnect domain account; the password manager service running on 01 and stopped on 02; Network Level Authentication disabled on both unless GEHA Security has enabled it under Input 9; Remote Desktop Users holds only the connector's PSMConnect and PSMAdminConnect pair	BE	15
A-08	Image and hardening	1. Run az vm show for each virtual machine and record storageProfile.imageReference. 2. On each connector run winver and confirm the operating system. 3. Open Local Security Policy, AppLocker, and confirm rules exist for executables and DLLs. 4. Read the RDS SecurityLayer and MinEncryptionLevel registry values. 5. Confirm SMB1 is disabled.	Command output and screenshots	Both virtual machines built from the GEHA CIS Level 2 Windows Server 2025 image; AppLocker rules present from the CyberArk GPO; SecurityLayer and MinEncryptionLevel set by the GPO; SMB1 disabled	BE	15
A-09	Probe health after hardening	1. After A-08, open the load balancer backend pool health. 2. If a probe reads unhealthy, inspect the Windows firewall rules the CyberArk GPO applied for inbound 3389 and 443 from 168.63.129.16.	Screenshot of backend health	Both servers Healthy on both probes. If a GPO rule blocks the probe source, add an inbound exception for 168.63.129.16 on the connector and record it as a GPO customization to re-apply after upgrades	BE	10
A-10	Health endpoint and DNS	1. Run nslookup cyberark-prd.geha.com from a client in the subnet and from a workstation in the administrative ranges. 2. Request https://CONNECTOR/psm/api/health on each server, substituting the connector name. 3. Request the same path through cyberark-prd.geha.com.	Command output; response captures	The name resolves to 10.9.250.7 (or the confirmed front end) from both places; HTTP 200 from each server and through the load balancer name	BE	10
A-11	Certificate binding	1. In IIS on each connector open the site binding for 443 and export the certificate details. 2. From a client in the administrative ranges, browse to https://cyberark-prd.geha.com/psm/api/health and inspect the certificate.	Certificate details and browser capture	Subject alternative names include the server name and cyberark-prd.geha.com; issued under the GEHA certificate standard; chain trusted; not expired; renewal date recorded for Venafi	BE	10
A-12	Tags	1. Run az resource list with tag output for rg-sec-cyberark-prd.	Export	Every resource carries the keys and values of the build tables tag set, including the backup exclusion tag	BE	5
A-13	Endpoint protection	1. On each connector record the endpoint protection agent installed, its state and its last check-in with the console. 2. Confirm the CyberArk exclusions are applied. 3. Confirm the agent was re-enabled after the connector install.	Screenshots; console record	The agent GEHA Security named under build tables input 22 is installed and enabled on both with the CyberArk exclusions; both servers present in its console	BE, SO	15

Group A, 13 cases, about 2.8 hours of execution..

Group B: Network path verification

Day 1, afternoon. One case per rule in the build tables firewall rule set. Run from each connector unless the step says otherwise. Every path crosses the hub firewall; there is no network security group, and traffic between subnets inside vnet-cip-cus-prd-01 is not filtered.

Ref	Case	Steps	Evidence	Expected result	Role	Min
B-01	Vault protocol	1. On each connector run Test-NetConnection VAULTHOST -Port 1858, substituting the vault host from the tenant export. 2. GEHA Network pulls the hub firewall log entry for the connection.	Command output; firewall log line	TcpTestSucceeded True on both; the log shows allow-vault-1858 matched on the exact host name	BE, NW	10
B-02	Tenant and connector management services	1. On each connector run Test-NetConnection on port 443 to one host in each FQDN set of allow-cyberark-services and allow-connector-management. 2. Run the same test to a public host in no set. 3. Run the revocation hosts on port 80.	Command output for every host	Every listed host succeeds; the unlisted host is refused; the firewall log shows the application rule matched for each success	BE, NW	20
B-03	Rapid7 agent	1. On each connector confirm the Insight agent service is running. 2. In the Insight console locate both servers and record the last check in.	Service screenshot; console screenshot	Both servers registered; last check in within 15 minutes; allow-rapid7 matched in the log	BE, SO	10
B-04	Domain controllers	1. On each connector run klist purge, then klist get krbtgt. 2. Run nslookup geha.com and nslookup of the vault host. 3. Run w32tm /resync.	Command output	Ticket granting ticket issued; DNS answers from a GEHA.com domain controller; time resync succeeds; allow-directory matched	BE	10
B-05	Splunk forwarding	1. On each connector run splunk list forward-server. 2. In Splunk search index=* host=CONNECTOR earliest=-15m. 3. In the deployment server console confirm both servers appear.	Command output; search results; console screenshot	Forwarder connected to an indexer on 9998; events from both servers present; both listed under spldepprd101 on 8089	BE, SO	15
B-06	MECM client	1. Push the client from MECMSITEPRD101 to both connectors. 2. In the console open the device record for each. 3. On each connector open Configuration Manager in Control Panel and run Machine Policy Retrieval. 4. Confirm client notification on TCP 10123 in the client log.	Console screenshots; client log extract	Client installed over allow-sccm-mid-inbound; device record active with a management point assignment; policy retrieval and client notification succeed over allow-sccm-client; both devices in the connector maintenance window collection	SO	30
B-07	ServiceNow discovery	1. Run a discovery schedule from SNMIDPRD301 that includes the connector subnet. 2. Open the resulting CI records.	Discovery status; CI screenshots	A CI exists for each connector with the correct name, addresses and operating system	SO	20
B-08	Remote desktop licensing	1. On each connector run Get-RDLicenseConfiguration. 2. Open Event Viewer, TerminalServices-Licensing, after the first brokered session in D-01.	Command output; event log extract	RDSLICPRD301 listed as the license server; a client access license issued on first session over allow-rds-licensing	BE, SO	10
B-09	Administrator entry	1. From a workstation in the GEHA administrative ranges, RDP to each connector name. 2. From an on-premises workstation outside those ranges, repeat.	Connection results	Succeeds from the administrative ranges (allow-admin-rdp); refused from outside them, with the firewall log showing no matching rule	TU, NW	15
B-10	Session path from each client group	1. RDP to cyberark-prd.geha.com from an on-premises workstation, from an Azure Virtual Desktop session host and from a Zscaler private access client. 2. GEHA Network pulls the log line for each.	Connection results; firewall log lines	All three reach the load balancer front end; each log line shows allow-session-from-clients matched with the expected source group	TU, NW	20
B-11	Inbound from another landing zone network	1. From a server in a landing zone network other than vnet-cip-cus-prd-01, run Test-NetConnection CONNECTOR -Port 3389. 2. From a server in another subnet of vnet-cip-cus-prd-01, repeat and record the result.	Command output	Refused from the other network, with no matching rule in the log. The result from inside vnet-cip-cus-prd-01 is recorded as is; traffic inside the network is not filtered (build tables input 20)	BE, NW	10
B-12	Platform endpoints	1. On each connector run Test-NetConnection on port 443 to a Windows Update host, a Defender for Endpoint host and an Azure Monitor host as published by Microsoft.	Command output	All succeed, through app-cyberark-platform or the existing policy rules GEHA Network named under Input 11	BE, NW	10

Group B, 12 cases, about 3 hours of execution..

Group C: Tenant registration

Day 1, end of day. The vendor verification steps for a new connector, applied to both servers.

Ref	Case	Steps	Evidence	Expected result	Role	Min
C-01	Connector Management status	1. In the Privilege Cloud portal open Connector Management, Connectors. 2. Select the row for each Azure connector and open the component list. 3. Record the status and version of every component. 4. Repeat for PASPCCPRD101 and PASPCCPRD102.	Screenshots of each component list	Every component on both Azure connectors shows an active status with no failed component; both on-premises connectors unchanged	CA	15
C-02	System health	1. Open the system health dashboard in the portal. 2. Record the entries for both Azure connectors.	Screenshot	Both Azure connectors present and healthy	CA	5
C-03	Identity Connector	1. On each connector open the Identity Connector Configuration program, Status tab. 2. Record the last connection time and result, and the auto-update setting and window.	Screenshots	Last connection recent and successful on both; both registered under the same customer ID; auto-update set as GEHA decided under Input 26	CA	10
C-04	Session manager service and farm entry	1. On each connector confirm the CyberArk Privileged Session Manager service is running. 2. In the portal open Configured PSM Servers and record the Azure-PSM-PRD entry and the on-premises entry.	Screenshots	Service running on both; the Azure entry shows ID Azure-PSM-PRD, Address cyberark-prd.geha.com and the PSM credential object; the on-premises entry unchanged	CA	10
C-05	Dynamic access connector	1. In the portal open the dynamic access connector pool and record both connectors.	Screenshot	Both connectors registered and online in the designed pool	CA	5
C-06	Password manager	1. In Connector Management record the password manager component on each Azure connector and its state. 2. In the portal confirm AZ-CPM-PRD-01 is listed.	Screenshots	Component present on both; running on connector 01 as AZ-CPM-PRD-01; installed and not started on connector 02 (the passive instance). If GEHA Security rules otherwise under Input 7, the recorded state is the ruled one	CA	10
C-07	Management Agent upgrade window	1. In Connector Management open the automatic upgrade setting for the Management Agent and record whether it is on, and the window.	Screenshot	Set as GEHA decided under Input 26: the CAB maintenance window, or opted out	CA	5

Group C, 7 cases, about 1 hour of execution..

Group D: Brokered sessions to migrated targets

Days 2 and 3. Proves the Azure farm serves a migrated server end to end, distributes across both connectors and survives the loss of one.

Ref	Case	Steps	Evidence	Expected result	Role	Min
D-01	RDP session through the Azure farm	1. Test user signs in to the portal from the administrative ranges. 2. Opens the Windows test account (test platform carrying the Azure farm ID) and clicks Connect. 3. Runs one command on the target that leaves a trace (for example creates a file). 4. Ends the session. 5. Reviewer opens the session in Monitoring and records the PSM server name.	Session record with PSM server name; recording; target evidence	Session brokered by AZPASPCCPRD101 or 102 over allow-brokered-and-rotation; recording complete and playable; the trace exists on the target	TU, CA	20
D-02	SSH session through the Azure farm	1. As D-01 with the Linux test account.	As D-01	As D-01 on TCP 22	TU, CA	15
D-03	Distribution across connectors	1. Open ten sessions in succession to the Windows test target, closing each before the next. 2. Record the PSM server for each session.	Session list export	Sessions land on both connectors; no run of sessions pinned to one server. The distribution is by hash; the result is recorded against Input 8	TU, CA	30
D-04	Connector failure	1. Stop the CyberArk Privileged Session Manager service on connector 01. 2. Note the time; watch the load balancer backend health. 3. Open three sessions. 4. Start the service; watch backend health return.	Timestamps; backend health screenshots; session records	Connector 01 marked unhealthy within 90 seconds through probe-psm-health; all three sessions brokered by 02 with no user error; 01 returns to Healthy after restart	BE, TU	30
D-05	Zone loss	1. Deallocate vm-sec-cyberark-cus-prd-01 (zone 1). 2. Repeat D-01 and D-06. 3. Start the virtual machine and confirm C-01 status returns.	Portal screenshots; session record	Sessions continue on connector 02; recordings retrievable; 01 rejoins with all components active	BE, TU, CA	40
D-06	Recording retrieval	1. Reviewer retrieves the D-01 and D-02 recordings from the portal and plays each to the end.	Playback screenshots	Both recordings complete, with the session end visible	CA	10
D-07	Live session monitoring	1. Test user opens a session as in D-01 and keeps it open. 2. Reviewer opens	Screenshots	Monitoring connects to the connector running the	TU, CA	20

Ref	Case	Steps	Evidence	Expected result	Role	Min
		live monitoring of that session from the portal. 3. Record whether monitoring connects and to which server.		session. If it fails, record it against Input 8: CyberArk routes live monitoring to the server that holds the session, which the hash-distributed load balancer cannot guarantee		
D-08	Concurrent sessions	1. Five test users, or one user with five accounts, open sessions in parallel to the test targets and hold them for ten minutes. 2. Record CPU and memory on both connectors from Azure Monitor.	Session list; metrics charts	All sessions brokered; CPU and memory within headroom for Standard_D8ls_v5, recorded against Input 6	TU, BE	30
D-09	Password manager verification	1. Reassign the test safe to AZ-CPM-PRD-01. 2. In the portal open the Windows test account, Overview, Last Verified, and click Verify. 3. Wait one CPM cycle. 4. Record the compliance status and which CPM performed the verification.	Screenshots	Account verified by AZ-CPM-PRD-01 from connector 01 over allow-brokered-and-rotation	CA	25
D-10	Account discovery	1. Run a discovery that includes the migrated test target. 2. Confirm the target is found.	Discovery results	Target discovered through the tenant scanner; no dependency on the Azure connectors	CA	15

Group D, 10 cases, about 3.9 hours of execution..

Group E: Parallel run and separation

Day 3. The controls that make the parallel run safe: the on-premises estate is unchanged, and no session can cross from the Azure farm to an unmigrated server.

Ref	Case	Steps	Evidence	Expected result	Role	Min
E-01	On-premises estate unchanged	1. Test user opens a session to the unmigrated test target (account on a production platform carrying the on-premises farm ID). 2. Reviewer records the PSM server.	Session record	Brokered by PASPCCPRD101 or 102; recording playable	TU, CA	15
E-02	No session from Azure to an unmigrated server	1. CyberArk administrator moves the unmigrated target account to the test platform carrying the Azure farm ID. 2. Wait one configuration refresh (ConfigurationRefreshInterval, default 600 seconds). 3. Test user attempts a session. 4. Administrator moves the account back and waits one refresh. 5. Repeat E-01.	Session attempt result; firewall log; E-01 repeat record	The attempt fails to reach the target; the hub firewall log shows deny-brokered-to-onprem matched; E-01 succeeds again after the move back	CA, TU, NW	40
E-03	Platform re-point rehearsal	1. Move the migrated Windows test account from the test platform (Azure ID) to a platform carrying the on-premises ID. 2. Wait one refresh; open a session; record the PSM server. 3. Move it back; wait one refresh; open a session; record the PSM server.	Two session records; account property export before and after	Sessions follow the platform on the next connection; no account property lost	CA, TU	40
E-04	Secure Tunnel and audit feed unaffected	1. On PASPCCPRD101 confirm the Secure Tunnel service is running. 2. In the SIEM confirm audit events from the tenant are still arriving.	Service screenshot; SIEM search	Running; events arriving. The re-point of this feed is build tables input 18 and is not part of this test	CA, SO	10
E-05	Credential provider consumers remain on premises	1. Pull the IIS logs from the four PASCCP hosts for the test window. 2. Filter for source addresses in the migrated test target set.	Log extract	No requests from any migrated test target. Any request found is recorded against Input 19, since the credential provider tier stays on premises with its consumers	CA	15

Group E, 5 cases, about 2 hours of execution..

Group F: Operations evidence

Day 3. Confirms the platform services that run the servers see them and that the operating arrangements in the build tables are in place.

Ref	Case	Steps	Evidence	Expected result	Role	Min
F-01	Backup protection and restore	1. In the Cohesity console open the protection status of both virtual machines and the policy assigned. 2. Confirm a snapshot completed after build. 3. Restore one operating system disk to an isolated resource group, or schedule the restore and record the date.	Console screenshots; restore job record	Both protected by the policy confirmed under Input 25 (the 8-week policy of the on-premises connector); snapshot present; restore succeeds or is scheduled; no Azure Backup enrollment	SO	30

Ref	Case	Steps	Evidence	Expected result	Role	Min
F-02	Alerting	1. Stop the PSM service on one connector for two minutes. 2. Record the alert received in Splunk and its route.	Alert record	The Splunk alert the Splunk owner configured for the connectors fires and reaches the named route	BE, SO	15
F-03	Patching	1. Deploy one approved update to both connectors through MECM in the connector maintenance window collection, connector 02 first and connector 01 the following night. 2. Confirm installation and reboot of each. 3. Confirm sessions continued on the other connector during each reboot.	MECM deployment status; session records	Installed on both; reboots completed one at a time; C-01 status active afterwards; no session interruption on the connector not being patched	SO, TU	40
F-04	Agent upgrade windows	1. Record the Management Agent automatic upgrade setting and window (C-07) and the Identity Connector auto-update window (C-03) against the CAB maintenance window.	Screenshots	Both set as GEHA decided under Input 26; neither can start outside the agreed window unless opted out	CA	10

Group F, 4 cases, about 1.6 hours of execution..

Group G: Disaster recovery

Day 4, after day 3 closes and before hypercare exit. Proves the connectors replicate to East US 2 and start there. The test failover uses an isolated virtual network so that production, the DR subnet and the tenant registration of the production pair are untouched; G-09 proves production is unaffected while it runs. The names used are those of the disaster recovery ruling; the cases run against the disaster recovery edition of the build tables when it is issued.

Ref	Case	Steps	Evidence	Expected result	Role	Min
G-01	Replication status	1. In the portal open the Recovery Services vault rsv-sec-cyberark-eus2-dr-01 in rg-sec-cyberark-dr, Replicated items, and record both connectors with status, RPO and last recovery point. 2. Record the time from enable replication to Protected.	Screenshots; timing note	Both items Protected; RPO within the replication policy; recovery points present for both	BE, CP	15
G-02	Replication paths	1. On each connector run Test-NetConnection on port 443 to the cache storage account blob endpoint, to login.microsoftonline.com and to the Site Recovery service host for the vault region.	Command output	All succeed on both connectors through the Site Recovery rules of the disaster recovery edition	BE, NW	15
G-03	Test failover	1. Open the recovery plan and run Test failover with the latest processed recovery point into the isolated test-failover virtual network. 2. Record start and completion times. 3. Record both test virtual machines with zone, size and tags.	Job record; screenshots; timing note	Both replicas running in East US 2, zone 1 and zone 2, Standard_D8ls_v5; the plan completes without error; tags recorded, including the backup exclusion tag	BE, CA	60
G-04	Replica start and registration	1. From a jump host in the isolated network, on each test virtual machine record the computer name, domain membership and the state of the CyberArk Privileged Session Manager service. 2. In Connector Management record what the tenant shows for the replica while it runs.	Screenshots	Computer names AZPASPCCPRD101 and 102; PSM service running. The tenant behaviour for a second instance of a registered connector is recorded, whatever it is; a conflict or duplicate entry becomes a recovery-plan action owned by GEHA Security	CA, BE	30
G-05	DR load balancer	1. Open the DR load balancer and record the front end, probes and rule. 2. Add the two test virtual machine NICs to the backend pool and record health. 3. Remove them after G-08.	Screenshots	Parameters match the build tables load balancer parameters for the DR names; both test virtual machines Healthy on both probes	BE	20
G-06	DNS re-point rehearsal	1. GEHA DNS creates cyberark-drtest.geha.com pointing at the DR front end address. 2. From the isolated network run nslookup and request https://cyberark-drtest.geha.com/psm/api/health. 3. Record the time to change and propagate; remove the record after G-08.	Command output; timing note	The record resolves; HTTP 200 through the DR load balancer; change time recorded	CP, SO	30
G-07	Tenant allowlist	1. GEHA Network provides the East US 2 hub egress public address. 2. CyberArk administrator confirms it is on the tenant allowlist.	Screenshot	Address present on the allowlist before any real failover	CA, NW	15
G-08	Cleanup	1. Run Cleanup test failover on the recovery plan with notes. 2. Confirm the test virtual machines are removed and replication remains Protected.	Job record; screenshot	Test virtual machines removed; both items Protected	BE	20
G-09	Production unaffected	1. During G-03 and after G-08 repeat D-01 against production. 2. Reviewer records the PSM server and the Connector Management status of the production pair.	Session records; screenshots	Brokered by AZPASPCCPRD101 or 102 both times; the production pair active throughout	TU, CA	20

Group G, 9 cases, about 3.8 hours of execution, plus the initial replication before day 4..

Schedule

The cases total about 14 hours of execution across the three test days and about 3.8 hours on day 4: groups A, B and C on day 1, group D on days 2 and 3, groups E and F on day 3, group G on day 4. Waiting time for configuration refresh (600 seconds per platform change), password manager cycles, initial replication and service owner actions is additional. Day 3 closes with the evidence pack and the summary sheet for the Security and Enterprise Architecture presentation; day 4 evidence is added to the pack before hypercare exit.

Exit criteria

Ref	Criterion
X-01	Groups A, B and C pass, or each failure has a named owner and a fix date before the CAB agenda request.
X-02	D-01 to D-07 and D-09 pass. D-08 passes or the sizing is re-examined before cutover.
X-03	E-01, E-02 and E-04 pass with no exception.
X-04	F-01 shows both connectors protected under the confirmed Cohesity policy and outside Azure Backup; F-03 shows one-at-a-time patching with no session interruption.
X-05	Open inputs that shape the plan are recorded with an owner and a date: password manager placement, load balancing method and live monitoring route, Secure Tunnel audit feed re-point, credential provider disposition, agent upgrade windows.
X-06	The evidence pack is complete: one subfolder per case, each holding the evidence named in the case, and a signed summary sheet.
X-07	Before hypercare exit: G-01, G-03, G-07, G-08 and G-09 pass; the G-04 result is recorded with any recovery-plan action and its owner; the G-06 change time is recorded.

Exit criteria.

Cutover day validation and rollback

Cutover for this application is the first production platform, or set of production accounts, pointed at the Azure farm. The steps run on the cutover day in order; CV-06 runs only if rollback is invoked. Cutover is approved by GEHA Security, Enterprise Architecture and CAB together and is scheduled in a CAB maintenance window, Thursday evening or a weekend, outside the freeze periods in the wave plan; the rollback decision belongs to Phil Costanzo, application admin and support manager, on the recommendation of the CyberArk administrator. The CyberArk PAS Rollback Plan holds the full procedure.

Ref	Step	Detail	Role	Min
CV-01	Pre-cutover check	Morning of cutover: repeat A-09, A-10, C-01 and C-02, and, once the disaster recovery edition is in service, confirm G-01 shows both replicated items Protected.	BE, CA	30
CV-02	Production re-point	Apply the Azure farm ID to the first production platform, or move the first production accounts to the Azure platform, and reassign their safes to AZ-CPM-PRD-01. Wait one configuration refresh.	CA	20
CV-03	Production session check	Repeat D-01 and D-02 against one production target per platform re-pointed.	TU, CA	30
CV-04	On-premises check	Repeat E-01 for one unmigrated production target.	TU, CA	15
CV-05	Both farms active	Confirm the tenant session list shows sessions on both the Azure and the on-premises farms.	CA	10
CV-06	Rollback, if invoked	Restore the on-premises farm ID on the platform, or move the accounts back, and reassign the safes to the on-premises password manager. Wait one refresh. Repeat CV-03 against the on-premises farm. No infrastructure change is needed.	CA	30

Cutover day steps.

Open inputs that shape this plan

The following are owed by GEHA in the build tables inputs table and change how a case is read rather than whether it runs: the password manager placement (Input 7, cases C-06 and D-09); the load balancing method and a direct route for live session monitoring (Input 8, cases D-03 and D-07); the Secure Tunnel audit feed re-point (Input 18, case E-

04); confirmation that the credential provider tier remains on premises with its consumers (Input 19, case E-05); the MECM collection, the Cohesity policy and the agent upgrade windows (Inputs 24 to 26, entry criterion E-11 and group F); and any recovery-plan action recorded at G-04. Their outcomes are recorded in the program decision register, not in this plan.

Workflow description

This describes the validation workflow drawing as swimlanes, phases, tasks, gates and decisions. Task identifiers are the case references; gate identifiers are new.

Swimlanes

Seven lanes, top to bottom: Cloud Platform Team (CP), GEHA Network (NW), Build engineer (BE), CyberArk administrator (CA), Test user (TU), Service owners (SO), Security and EA reviewers (SR). GEHA Identity's entry items sit in the build engineer lane. The evidence store is drawn as a data store at the right edge, receiving an artifact from every task.

Phases and gates

Phase	Lane	Tasks in order	Gate at the end	Gate rule
0 Entry	CP, NW, BE, CA, SO	E-01 to E-11 of the entry criteria	G0 Ready to test	All eleven entry criteria confirmed; otherwise return to the owner of the failed criterion
1 Platform build (day 1 am)	BE	A-01 to A-13, sequential; A-09 follows A-08 directly	G1 Built as specified	No failed case, or every failure has an owner and a fix date; A-09 unhealthy sends a Windows firewall exception task to BE before G1
2 Network paths (day 1 pm)	BE, NW, SO, TU	B-01 to B-12; B-03, B-05, B-06, B-07 may run in parallel with the service owners	G2 Paths open	All paths pass and the two deny checks refuse; a refused allow path returns to NW for the rule collection
3 Tenant registration (day 1 close)	CA	C-01 to C-07 in order	G3 Registered	C-01 to C-05 active on both connectors; C-06 shows the password manager active on 01 and passive on 02; C-07 recorded
4 Brokered sessions (days 2 and 3)	TU, CA, BE	D-01, D-02, D-06, D-03, D-04, D-05, D-07, D-08, D-09, D-10	G4 Serves migrated targets	D-01 to D-07 and D-09 pass; D-08 pass or sizing review; a D-07 failure is recorded against the load balancing input and a decision is taken before G4
5 Parallel run (day 3)	CA, TU, NW, SO	E-01, E-02, E-03, E-04, E-05	G5 Estates separated	E-01, E-02 and E-04 pass with no exception; an E-02 session that reaches the target is a stop: cutover cannot be requested
6 Operations (day 3)	SO, BE, CA	F-01, F-02, F-03, F-04	G6 Operable	F-01 protected and outside Azure Backup; F-03 one at a time with no interruption; F-02 and F-04 pass or deferred with owner
7 Acceptance	BE, SR	Assemble evidence pack and summary sheet; one hour presentation to Security and EA	G7 Accepted	Both teams accept; playbook status set to Finalized; CAB agenda request submitted the Friday before CAB
8 Cutover day	BE, CA, TU	CV-01 to CV-05; CV-06 on rollback	G8 Cut over	CV-03, CV-04 and CV-05 pass; otherwise CV-06 and a return to G7 for the next CAB
9 Disaster recovery (day 4, before hypercare exit)	BE, CP, NW, CA, TU, SO	E-12, then G-01, G-02, G-03, G-04, G-09 during G-03, G-05, G-06, G-07, G-08, G-09 after G-08	G9 Recoverable	G-01, G-03, G-07, G-08 and G-09 pass; G-04 recorded with any recovery-plan action and owner; G-06 change time recorded

Phases, tasks and gates.

Decision points

Decision	Where	Yes path	No path
Probes healthy after hardening?	After A-09	A-10	Add a Windows firewall exception for 168.63.129.16, record it as a GPO customization, rerun A-09
All tenant paths open?	After B-02	B-03	Return to NW: rule collection fix, rerun B-01 and B-02
Password manager active on 01, passive on 02?	C-06	D-09 verifies through AZ-CPM-PRD-01	Correct the service state or record the GEHA ruling under Input 7; rerun C-06

Decision	Where	Yes path	No path
Live monitoring reaches the host?	D-07	Continue	Record against Input 8; decide the direct monitoring route before G4
Session from Azure reaches an unmigrated server?	E-02	Stop: firewall finding to NW; no cutover request until rerun passes	Continue to E-03
Evidence accepted by Security and EA?	G7	Finalize playbook, submit CAB agenda request	Address findings, re-present
Production sessions pass on cutover day?	CV-03 to CV-05	Hypercare begins	CV-06 rollback; return to G7 for the next CAB
Replica registers with the tenant?	G-04	Continue to G-05	Record a re-registration step as a recovery-plan action owned by GEHA Security; continue to G-05
Test failover cleaned up and replication Protected?	G-08	G-09, then G9	Remove the test resources by hand, record, rerun G-01

Decision points.

Drawing notes

Draw phases as columns left to right and lanes as rows. Each task is a box labeled with its reference and title, placed in the lane of the role that runs it; tasks with two roles sit in the first role's lane with a dashed link to the second. Gates are diamonds on the phase boundary spanning all lanes, labeled G0 to G9. Decision points are diamonds inside a phase with Yes and No exits labeled per the decision points table. Every task has a thin arrow to the evidence store. Waiting time (configuration refresh, password manager cycle, initial replication) is drawn as a clock marker on E-02, E-03, D-09, CV-02 and G-01. The cutover day phase is separated from the test phases by a heavier vertical rule labeled CAB approval; the disaster recovery phase follows it and ends at a vertical rule labeled Hypercare exit.

Revision history

Version	Date	Change
v2.1	28-Sep-2026	Prepared for merge: section, table, figure and appendix numbering removed; references restated by name against Build Tables v2.3.
v2.0	28-Sep-2026	Rebuilt against Build Tables v2.2 Production and Handover v2.2. Network security group cases removed; every path case names its hub firewall rule; client-group session path (B-10), other-network deny (B-11) and platform endpoints (B-12) added. Image and hardening cases rewritten for the CIS Level 2 Windows Server 2025 image with the CyberArk GPO as the only hardening policy; endpoint protection case A-13, organizational unit check in A-06, DNS in A-10, backup enrollment in A-04. Password manager treated as active on 01 and passive on 02 (C-06, D-09); Management Agent and Identity Connector upgrade windows (C-03, C-07, F-04); one-at-a-time patching (F-03). Credential provider recorded as retained on premises. Roles split into Cloud Platform Team, GEHA Network and GEHA Identity. Entry criteria E-11 operations set up added; DR criterion renumbered E-12. Group G retained against the disaster recovery edition. Open items keyed to the build tables inputs.
v1.1	23-Sep-2026	Basis moved to Azure Build Parameters v3.2. Register codes removed. Group G, disaster recovery, added with entry criterion, exit criterion, phase 9 and gate G9. Workflow diagram added as Figure 1.
v1.0	15-Sep-2026	Initial plan: groups A to F, entry and exit criteria, cutover day steps, workflow description.

Revision history.